Saturday, 01 November 2025

EXAM POV NOTES (Q&A FORMAT)

10 Marks Questions

Q1. Explain the concept and necessity of Multiple Access Protocols in computer networks, providing the classroom analogy used in the source, and describing the condition under which they become required instead of just the data link control layer.

Multiple Access Protocols (MAPs) are essential methods used in networks without dedicated links to manage traffic flow, prevent collisions, and ensure the efficient sharing of the communication channel among multiple devices.

- **Definition of MAPs:** Multiple Access Protocols are methods used to control how data is transmitted when multiple devices are attempting to communicate over the same network.
- **Purpose (Efficiency):** These protocols ensure that data packets are sent and received efficiently, specifically to prevent collisions or interference.
- **Necessity Condition:** MAPs are required when there is **no dedicated link** present between the sender and the receiver, allowing multiple stations to access the channel simultaneously.
- **Data Link Sufficiency:** If a dedicated link *is* present between the sender and receiver, the data link control layer alone is sufficient for managing data transmission.
- **Traffic Management:** They help manage the network traffic so that all devices can share the communication channel smoothly and effectively.
- **Collision Mitigation:** The protocols are fundamentally required to decrease collision rates and to avoid crosstalk (unwanted signal interference).
- **The Classroom Analogy (Chaos):** If a teacher asks a question (channel access) and all students (stations) start answering simultaneously (sending data at the same time), chaos is created, resulting in data overlap or data loss.
- **The Classroom Analogy (Protocol Role):** In this scenario, it is the job of the "teacher" (representing the multiple access protocols) to manage the students and make them answer one at a time.
- **Protocol Requirement:** Thus, MAPs are explicitly required for sharing data effectively on non-dedicated channels.
- **Protocol Subdivisions:** These protocols are subdivided into categories, including ALOHA, Controlled Access, and Channelization methods.

Q2. Detail the operational mechanism of Carrier Sense Multiple Access with Collision Detection (CSMA/CD). Include the requirement for frame size, which dictates that the frame transmission delay must be at least two times the maximum propagation delay, and explain the concept of contention slots.

CSMA/CD enhances basic CSMA by adding a mechanism to detect collisions. This requires the frame transmission duration to be long enough to capture the collision signal before transmission ends, defining the minimum frame size requirement based on propagation delay and utilizing contention slots.

- **CSMA/CD Function:** This method adds to the basic CSMA algorithm by providing a specific way to deal with collisions that occur during transmission.
- **Collision Detection Goal:** The primary mechanism requires the size of a frame to be large enough so that the sender can successfully detect a collision while the frame is still being transmitted.
- **Minimum Transmission Delay Rule:** The required condition is that the frame transmission delay **must be at least two times the maximum propagation delay** ().
- **CSMA Pre-Check:** Like basic CSMA, the station is still required to first sense the medium (for idle or busy status) before it attempts to transmit data.

- **Worst Case Consideration:** The protocol must account for the **Worst Case scenario** when analyzing efficiency, even though successful transmission is the Best Case.
- **Definition of Contention Slots: Contention slots** are defined as those time slots during which data packets are not able to complete their journey because a collision has occurred.
- **Worst-Case Time Wasted:** If a station transmits data but collides, the maximum time that can be wasted in the worst case is defined as .
- **Detection Mechanism:** The collision detection process involves the sender monitoring for additional signals (its own signal plus the signal of the colliding station) to confirm a collision.
- **Immediate Re-transmission Avoidance:** After a collision, the system accounts for the wasted time () before another station finds a way to successfully transmit the data.
- **Necessity Over Basic CSMA:** CSMA/CD is necessary because the basic CSMA method alone does not prescribe what action a station should take once a collision is detected.

Q3. Describe the comprehensive architecture of the Asynchronous Transfer Mode (ATM) protocol. Explain the specific roles and functions of the three main layers: the ATM Adaptation Layer (AAL), the ATM Layer, and the Physical Layer.

ATM utilizes a connection-oriented architecture designed to handle multiple service types using fixed-size 53-byte cells. Its protocol architecture is defined by three layers that manage data flow from user applications down to the physical transmission medium.

- **AAL Isolation Function:** The ATM Adaptation Layer (AAL) functions primarily to **isolate higher-layer protocols** (like voice, video, or data) from the precise details of the underlying ATM processes.
- **AAL Segmentation Role:** AAL accepts user data from higher layers and makes them suitable for ATM transmission by segmenting these packets into the fixed-sized 48-byte payloads required for ATM cells.
- **AAL Reassembly Role:** AAL is also responsible for the inverse process: reassembling the received cell segments at the destination back into the original higher-layer packets.
- **ATM Layer Core Functions:** The ATM Layer is responsible for crucial network management tasks, including transmission, switching, congestion control, and ensuring sequential delivery of cells.
- **ATM Layer Multiplexing:** This layer handles **cell multiplexing**, which is the process of simultaneously sharing the virtual circuits over the single physical link.
- **ATM Layer Relay:** It manages **cell relay**, which is the passing of cells through the ATM network, utilizing the VPI (Virtual Path Identifier) and VCI (Virtual Channel Identifier) information contained within the cell header.
- **Physical Layer Components:** The Physical Layer manages the medium-dependent transmission and is sub-divided into two parts: the Physical Medium-Dependent sublayer and the Transmission Convergence sublayer.
- **Physical Layer Bit Control:** The physical layer functions include controlling the transmission and receipt of individual bits across the physical medium.
- **Physical Layer Conversion:** It converts the cells received from the ATM layer into a continuous bitstream for transmission.
- **Physical Layer Framing/Tracking:** Key functions include tracking the precise ATM cell boundaries and looking after the packaging of cells into the appropriate type of transmission frames.

Q4. Compare and contrast the three modes of the VLAN Trunking Protocol (VTP): Server, Client, and Transparent. Your explanation must differentiate them based on their ability to create/delete VLANs, their role in propagating updates, and whether they save configurations in NVRAM.

VTP is a Cisco proprietary protocol designed to synchronize VLAN configuration data across a VTP domain. The three modes dictate the specific configuration rights and update propagation behavior of the switch.

- **Server Mode (Default):** Switches are set to Server mode by default. This is where all configuration changes should be performed.
- **Server Capabilities:** Server mode allows the user to **create, add, and delete VLANs**.
- **Server Propagation & Saving:** Any changes made in Server mode will be **advertised** to all other switches in the same VTP domain. The configurations are **saved persistently in NVRAM**.
- **Client Mode Capabilities:** Client mode switches **cannot create, add, or delete VLANs**; they only learn configuration updates.
- **Client Propagation & Saving:** Client switches receive updates from servers and can also **forward (propagate) those updates** to other switches in the domain. However, the received configurations are **not saved in NVRAM** and will be deleted if the switch is reset or reloaded.
- **Transparent Mode Role:** The main purpose of the Transparent mode is to forward the VTP summary advertisements through the trunk link, but the switch does not take part in the VLAN assignment synchronization process.
- **Transparent Capabilities:** Transparent mode switches **can make their own local VLAN database**, but this database is kept secret and is not shared or synchronized with other switches in the domain.
- **Transparent Propagation:** A Transparent switch only forwards the summary advertisements; it does **not process or apply** updates it receives, nor does it advertise its local changes.
- **Synchronization Dependency:** VTP relies on at least one switch being configured as a Server for synchronization to occur throughout the domain.
- **Update Flow:** All modes that are part of the domain (Server and Client) must have matching VTP domain names and VTP versions to communicate VLAN information successfully.

Q5. Explain the different Channelization Multiple Access Protocols: Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), and Code Division Multiple Access (CDMA). For each, describe the method used to share bandwidth and note the unique characteristic of CDMA that allows simultaneous transmission.

Channelization protocols enable multiple stations to access a channel simultaneously by dividing the available bandwidth into separate resources based on frequency, time slots, or unique code languages.

- **Channelization Principle:** In Channelization, the available bandwidth of the link is shared among multiple stations by dividing it based on time, frequency, or code, allowing simultaneous access.
- **FDMA Bandwidth Division:** Frequency Division Multiple Access (FDMA) shares bandwidth by dividing the available frequency spectrum into equal, non-overlapping bands.
- **FDMA Allocation:** Each individual station is allocated its own dedicated frequency band for communication.
- **FDMA Guard Bands:** Guard bands are added between the frequency bands to ensure that no two adjacent bands overlap, which prevents crosstalk and noise.

- **TDMA Time Division:** Time Division Multiple Access (TDMA) shares the entire bandwidth, but collision is avoided by dividing time into discrete slots.
- **TDMA Allocation:** Stations are allotted specific, alternating time slots during which they are permitted to transmit data.
- **TDMA Synchronization:** A key issue in TDMA is the overhead of synchronization, as every station must know its exact time slot. This is resolved by adding synchronization bits to each time slot.
- **CDMA Unique Characteristic:** Code Division Multiple Access (CDMA) is unique because there is neither a division of bandwidth nor a division of time; all transmissions happen simultaneously on one channel.
- **CDMA Sharing Method:** Data from different stations is transmitted simultaneously using **different code languages**.
- **CDMA Reception Principle:** Perfect reception is possible if only the intended receiver "speaks the same language" (uses the same code) as the sender, despite many others speaking at the same time on the shared channel.

5 Marks Questions

Q6. Explain the difference between Pure ALOHA and Slotted ALOHA. Describe the mechanism used in Pure ALOHA (involving the acknowledgement, back-off time, and re-sending) to decrease the probability of collision.

ALOHA protocols allow instant transmission, risking collision. Pure ALOHA uses a random delay mechanism for collision recovery, while Slotted ALOHA imposes strict time boundaries to prevent initial collision.

- **Pure ALOHA Mechanism (Acknowledgement):** When a station sends data in Pure ALOHA, it immediately waits for an acknowledgement from the receiver.
- **Pure ALOHA Mechanism (Collision/Delay):** If the acknowledgement is not received within the allotted time (implying a collision occurred), the station waits for a random amount of time known as the **back-off time ()**.
- **Pure ALOHA Mechanism (Re-send):** After the random back-off time expires, the station then re-sends the data.
- **Collision Probability Decrease:** Because different stations use different random amounts of wait time, the probability of the stations colliding again on the subsequent attempt is decreased.
- **Slotted ALOHA Difference:** Slotted ALOHA divides time into discrete slots, and stations are permitted to send data **only at the beginning of these slots**. If a station misses the start time, it must wait for the next slot, which reduces the probability of collision compared to Pure ALOHA.

Q7. Define the four main CSMA Access Modes: 1-Persistent, Non-Persistent, P-Persistent, and O-Persistent, describing how each mode handles sensing a busy channel.

The CSMA Access Modes determine the station's behavior and persistence level when the channel is sensed as busy prior to transmission.

- **1-Persistent:** If the medium is sensed busy, the node continuously keeps checking the medium. As soon as the channel becomes idle, the node transmits the data unconditionally (with a probability of 1).

- **Non-Persistent:** If the medium is sensed busy, the node does **not** continuously check the channel. Instead, it checks the medium again only after a random amount of time has elapsed, transmitting when it eventually finds the channel idle.
- **P-Persistent (Probability):** If the medium is sensed idle, the node transmits the data with a specific probability.
- **P-Persistent (Wait):** If the data is not transmitted (with probability), the station waits for some time, checks the medium again, and transmits with probability . This probabilistic loop continues until the frame is sent.
- **O-Persistent:** This mode uses predetermined superiority (priority) decided beforehand. If the medium is idle, the node still waits for its specific time slot, and transmission occurs strictly in that defined order.

Q8. Describe the two specific mechanisms used by CSMA/CA to avoid collisions: Interframe Space (IFS) and the Contention Window.

CSMA/CA (Collision Avoidance) uses timing and random back-off to minimize collisions, unlike CSMA/CD which detects them after they start. The key methods are IFS and the Contention Window.

- **Interframe Space (IFS) Purpose:** The IFS mechanism requires a station, upon finding the medium idle, to wait for an additional period of time (Interframe Space) rather than sending immediately. This delay helps avoid collisions caused by propagation delay.
- **IFS Mechanism:** After the IFS duration, the station checks the medium a second time for idleness before proceeding with transmission.
- **IFS Duration:** The specific duration of the Interframe Space is determined by the priority level assigned to the station.
- **Contention Window Structure:** The Contention Window is a defined amount of time that is logically divided into sequential slots.
- **Contention Window Mechanism:** A sender ready to transmit chooses a **random number of slots** within the Contention Window as its wait time. If the medium is found busy, the timer restarts when the channel is next found idle, rather than restarting the entire process.

Q9. List and briefly explain the main components of the Point-to-Point Protocol (PPP), which are the Encapsulation Component, Link Control Protocol (LCP), Authentication Protocols (AP), and Network Control Protocols (NCPs).

PPP is a data link layer protocol used for transmitting multiprotocol data between two directly connected systems, organized around specific components for framing, link management, authentication, and network layer negotiation.

- **Encapsulation Component:** This component's function is to encapsulate the datagram (data packet) so that it can be correctly transmitted over the specified physical layer.
- **Link Control Protocol (LCP):** LCP is essential for managing the connection link, responsible for establishing, configuring, testing, maintaining, and terminating the links for transmission.
- **Authentication Protocols (AP):** These protocols are used to authenticate the two endpoints of the connection, verifying their identity for the use of services. The two types are Password Authentication Protocol (PAP) and Challenge Handshake Authentication Protocol (CHAP).
- **Network Control Protocols (NCPs):** NCPs are used to negotiate the parameters and facilities required for the network layer (Layer 3).
- **NCP Examples:** There is one NCP for every higher-layer protocol supported by PPP. Examples include the Internet Protocol Control Protocol (IPCP) and IPv6 Control Protocol (IPV6CP).

Q10. Discuss the essential services provided by Wireless Routers when used as integral parts of a Converged WAN. Mention their roles regarding wireless connectivity, hybrid WANs, and failover/load balancing.

Wireless routers are crucial components in a Converged WAN, integrating various traffic types (data, voice, video) over a unified infrastructure and providing resilience and seamless access.

- **Wireless Connectivity Provision:** Wireless routers provide seamless connectivity, which allows remote users and mobile devices to access the core WAN services without requiring physical cable connections.
- **Hybrid WAN Integration:** They enable the successful integration of wireless networks (such as 4G or 5G) with traditional WAN technologies (like MPLS or Ethernet) to create a **Hybrid WAN**.
- **Backup and Failover:** Wireless routers are utilized in converged WANs to serve as dedicated backup connections for primary wired links, thereby ensuring a failover mechanism in case of primary link failure.
- **Load Balancing:** They are capable of distributing traffic effectively across both the available wireless links and the wired links.
- **Bandwidth Optimization/Availability:** The combined roles of failover and load balancing help to optimize bandwidth usage and ensure high availability of services across the entire network.

Q11. Explain how Frame Relay operates to connect multiple Local Area Networks (LANs) to build a Wide Area Network (WAN). Specifically, detail the function of the Data Link Connection Identifier (DLCI) in identifying the destination LAN.

Frame Relay is a packet-switching WAN protocol that connects multiple LANs by setting up virtual circuits over shared access links. The DLCI is the essential identifier used by switches to route frames through these circuits.

- **WAN Setup using Switches:** Frame relay switches establish **virtual circuits** (logical paths) to link multiple LANs together, effectively constructing a Wide Area Network (WAN).
- **Access Links:** Each LAN connects its border device (e.g., a router) to the service provider network via a private physical link known as the **access link**, which terminates at a frame relay switch.
- **Frame Transmission:** Data is transferred by dividing it into packets (frames), which are transmitted across the shared physical links.
- **DLCI Examination:** When a LAN's router sends a data packet, the frame relay switch examines the packet specifically to retrieve the **Data Link Connection Identifier (DLCI)**.
- **DLCI Function:** The DLCI indicates the destination of the packet. The switch uses this identifier, combined with its existing address information, to pinpoint the destination LAN and transfer the packet to the destination LAN's frame relay switch.

Q12. Describe the concept of Virtual LANs (VLANs). Explain how switches use VLANs to divide the broadcast domain, which is generally handled by Layer 3 devices.

VLANs logically segment a network at Layer 2, allowing switches to create smaller, isolated broadcast domains, thereby improving manageability and control.

- **Logical Division at Layer 2:** A Virtual LAN (VLAN) is a concept that allows the logical division of devices at Layer 2 (the data link layer).
- **Broadcast Domain Definition:** A broadcast domain is a network segment where a broadcast packet sent by one device is received by every other device in that segment.

- **General Domain Division:** Typically, broadcast domains are divided by Layer 3 devices, such as routers, which do not forward broadcast packets.
- **Switch Division Capability:** Using the VLAN concept, Layer 2 devices (switches) are also able to divide the broadcast domain into smaller, separate segments.
- **Inter-VLAN Routing:** To enable communication between these separate VLANs (different broadcast domains), a mechanism known as **inter Vlan routing** is required.

Q13. Outline the core features and design elements of ATM technology. Focus on the structure of Cells (size and composition), and the role of Virtual Channels (VCs) and Virtual Paths (VPs) in connection establishment.

ATM technology is a high-performance networking standard based on connection-oriented cell switching, employing fixed-size data units and hierarchical virtual connections.

- **Cell Definition:** The basic data transmission units in ATM are called **cells**, which are small and fixed in size.
- **Cell Structure:** Each cell is precisely **53 bytes** long. This length is composed of a 5-byte header and a 48-byte payload.
- **Virtual Channels (VCs):** VCs (Virtual Channel Connections) are the most basic unit in ATM, responsible for carrying a single stream of cells from one user to another.
- **Virtual Paths (VPs):** VPs (Virtual Path Connections) consist of multiple VCs bundled together, sharing the same network path.
- **Connection Orientation:** ATM is a connection-oriented network. Establishing communication requires first sending a message to set up a connection, ensuring all subsequent cells follow that exact path.

Q14. State the primary characteristics and features of a Network Switch, noting its operation layer in the OSI Model and the types of communication it supports.

A Network Switch is a Layer 2 device that segments networks into subnets and intelligently forwards packets based on MAC addresses, supporting multiple communication modes.

- **OSI Layer Operation:** The switch operates at the **Data Link Layer** (Layer 2) of the OSI Model.
- **Core Function:** It is responsible for segmenting networks into smaller subnets and performs filtering and forwarding of packets based on the **MAC address**.
- **Communication Modes:** Switches support different types of communication modes: **unicast** (one-to-one), **multicast** (one-to-many), and **broadcast** (one-to-all) transmission.
- **Packet-Switching:** Switches use **packet-switching techniques** to transfer data packets reliably from the source to the destination device.
- **Error Checking:** Before forwarding data, the switch performs **error checking** to ensure data integrity.

Q15. Detail the operational implementation of the First Come First Serve (FCFS) CPU scheduling algorithm, including the use of the FIFO queue, and explain how the waiting time for an upcoming process is calculated.

FCFS is the simplest CPU scheduling algorithm, allocating the CPU based on arrival time and utilizing a non-preemptive FIFO queue structure.

- **Scheduling Principle:** FCFS schedules processes strictly according to their **arrival times**; the process that requests the CPU first is allocated the CPU first.
- **Queue Implementation:** The algorithm is implemented using the **FIFO (First-In, First-Out) queue** structure

- **Process Entry/Exit:** When a process enters the ready queue, its Process Control Block (PCB) is linked to the tail of the queue. When the CPU is free, it is allocated to the process at the head, which is then removed.
- **Non-Preemptive Nature:** FCFS is typically described as a **non-preemptive** scheduling algorithm, meaning a running task continues until completion or blocks itself.
- **Waiting Time Formula:** The waiting time () for an upcoming process is calculated using the arrival time and burst time of the previous process.

2 Marks Questions

Q16. State the four core requirements listed in the sources that must be met for VTP to successfully communicate VLAN information between switches.

- **VTP Version and Domain Match:** The VTP version must be the same on the switches, and the VTP domain name must also be the same across the configured switches.
- **Server and Authentication:** One of the switches must be designated as a server, and the authentication password must match across all communicating switches if authentication has been applied.

Q17. What are the two Authentication Protocols (AP) used by the Point-to-Point Protocol (PPP)?

- **Password Authentication Protocol (PAP):** One of the authentication protocols used by PPP endpoints to verify service usage.
- **Challenge Handshake Authentication Protocol (CHAP):** The second authentication protocol used by PPP.

Q18. In basic CSMA, why is there still a chance of collision even after a station senses the medium to be idle?

- **Propagation Delay:** Collision is still possible in basic CSMA due to the inherent **propagation delay** of signals across the medium.
- **Simultaneous Transmission:** If Station A starts transmitting, Station B might sense the medium as idle before the first bit of Station A's data reaches it (due to delay) and consequently start transmitting simultaneously, leading to a collision.

Q19. What is the fundamental assumption of independence among predictors that makes the classification technique referred to as the Bayesian Algorithm 'Naïve'?

- **Independence Assumption:** The Bayesian Algorithm is defined as 'Naïve' because it is built upon an assumption of **independence among predictors** (features).
- **Unrelated Features:** The algorithm assumes that the presence of a specific feature in a class is **unrelated** to the presence of any other feature, even if those features are logically dependent on one another.

Q20. Which VLAN range is defined as the default VLAN for switches, and why can it not be deleted or edited?

- **Default VLAN: VLAN 1** is the default VLAN for switches, and by default, all switch ports belong to this VLAN.
- **Restriction:** VLAN 1 cannot be deleted or edited, although it remains available for use.

Q21. What are the reserved VLAN ranges (VLAN IDs) that cannot be seen or used?

- **Reserved IDs: VLAN 0 and VLAN 4095** are reserved VLAN IDs that cannot be seen or used by network administrators.
- **CISCO Defaults: VLANs 1002 through 1005** are also reserved, designated as CISCO defaults for FDDI and Token Rings, and these specific VLANs cannot be deleted.

Q22. What is the main purpose of Guard bands in Frequency Division Multiple Access (FDMA)?

- **Placement:** Guard bands are added into the allocated spectrum to separate the frequency bands given to different stations.
- **Purpose:** Their purpose is to ensure that no two frequency bands overlap, which prevents **crosstalk and noise** interference.

Q23. What are the two functions of the Spanning Tree Protocol (STP) algorithm in managing topology?

- **Loop-Free Topology:** STP functions by computing a **loop-free portion of the topology** (a spanning tree) by declaring certain redundant switch ports and placing them into a "blocking" state.
- **Automatic Recovery:** It automatically recovers from a switch failure that might partition the extended LAN by reconfiguring the spanning tree to utilize redundant paths if available.

Q24. How long is an ATM cell, and how is that length specifically divided between the header and the payload?

- **Total Length:** Each ATM cell is a fixed size of **53 bytes** long.
- **Division:** This length is divided into a **5-byte header** and a **48-byte payload**.

Q25. What was AAL Type 5 originally named, and name two areas where it is used?

- **Original Name:** AAL Type 5 was originally named the **Simple and Efficient Adaptation Layer (SEAL)**.
- **Usage Areas:** It is used in areas such as **Internet Protocol (IP) over ATM** and **Ethernet over ATM**.